THE CURE CENTER
FOR ULTRASOUND RESEARCH & EDUCATION

INTRODUCTION TO FETAL ECHOCARDIOGRAPHY SYLLABUS

COURSE CODE: PED 200

INTRODUCTION TO FETAL ECHOCARDIOGRAPHY

Instructor:
Telephone:
E-Mail:
Office Hours:
Mon, Wed, Fri: 12pm to 1pm & 5pm to 6pm
Tues: 9am – 5pm
Thurs: 11am – 5pm

Start Date:

Completion Date:

Days and Times:
Monday, Wednesday and Friday

Course Hours:
Morning Session:
8:00 am -10:00am &
10:10am – 12:00 pm
Afternoon Sessions:
1:00pm – 3:00pm &
3:10 pm – 5:00 pm

Class Location:
Classroom #:
Lab#:

Prerequisites:
BIO 100 Introduction to A&P
PAT 100 Patient Care
LAW 100 Medical Law & Ethics

Course Contact Hours: 16
Lecture: 8
Laboratory: 8
Course Delivery: Residential
Pre-requisite: None

Americans with Disabilities Act Guidelines:

Requests for accommodations should be submitted to the campus president. Faculty are not authorized to grant accommodations.

COURSE DESCRIPTION

This introductory level course is directed towards developing the principals, techniques, skills and fundamental knowledge of each student necessary to function as a qualified entry level diagnostic medical cardiac sonographer.
It introduces the foundations of diagnostic medical sonography as it pertains to the echocardiography images, including medical terminology related to the heart anatomy, physiology and pathology. Students will learn hemodynamic principals and scanning plane orientations (longitudinal, transverse and coronal) related to echocardiography.

Emphasis will be placed on the clinical applications of echocardiography, including basic patient care principals, professionalism, clinical signs and symptoms. Scanning protocols and techniques will be presented in a demonstrated lesson in conjunction with laboratory scanning sessions. The academic laboratory and clinical courses are integrated into the program for students to learn and practice different echocardiographic scanning protocols. Interpretation of normal and pathologic echocardiography patterns will be learned including the normal variants, laboratory tests and values. At the end of this curriculum, students will possess knowledge and skills necessary for employment in the healthcare system as cardiac sonographer.

Instructional Materials

1. Textbook
Title: Textbook of Diagnostic Sonography, 8th Edition (2 Volume Set)
Author(s): Sandra L. Hagen-Ansert, MS, RDMS, RDCS, FASE, FSDMS
Publisher: Mosby; 8th edition (May 3, 2017)
ISBN: 978-0323353755

2. Workbook
Title: Workbook for Diagnostic Sonography, 8th Edition
Author(s): Sandra L. Hagen-Ansert, MS, RDMS, RDCS, FASE, FSDMS
Publisher: Mosby; 8th edition (May 3, 2017)
ISBN: 978-0323441834

3. Reference Book
Title: Pocket Protocols for Sonography Scanning, 4th Edition
Author(s): Betty Bates-Tempkin, BA, RT®, RDMS
Publisher: Saunders; 4th edition (December 27, 2014)
ISBN: 978-1455773220

4. Textbook
Title: ASE Comprehensive Echocardiography, 2nd Ed.
Author(s): Lang Goldstein
Publisher: Elsevier
ISBN: 978-0323260114

5. Textbook
Title: Feigenbaum's Echocardiography, 8 ed
Author(s): William F. Armstrong, Thomas Ryan, M.D.
Publisher: LWW
ISBN: 978-1451194272

Equipment/Technology

- LCD Projector
- Laptop Computer
- Mindray Ultrasound Machine

Material

- Electronic and Printed Worksheets
- Textbooks
- Visual Aids: DVD Videos, Case Study Reviews, Demonstrations and Hands on Clinical Laboratory Practice.

During clinical laboratory sessions various techniques such as role playing and scanning exercises will be used.

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Grade Point Value

Grade	Percentile Range	Points
A+	97-100	4.00
A	93-96.99	3.85
A-	90. -92.99	3.67
B+	87-89.99	3.33
B	83-86.99	3.00
B-	80-82.99	2.67
C+	77-79.99	2.33
C	73-76.99	2.00
C-	70-72.99	1.67
D+	67-69.99	1.33
D	63-66.99	1.00
D-	60-62.99	.67
F	Below 60-00	0

Methods of Evaluation

Exams / Quizzes/ Midterm/ Final	40%
Projects/ Attendance Participation/ Homework Assignments	20%
Scan-out	40%
	100%

Teaching Strategies

A variety of teaching strategies may be utilized in this course, including but not limited to: lecture, discussion, written classroom exercises, visual aids, videos, case study reviews, group and/or individual projects, quizzes, exams, reading assignments, writing assignments, interactive exercises demonstrations and hands, internet research and hands on clinical laboratory practice.

Examination Standards

Students are expected to be honest in all academic work. Cheating of any kind is not tolerated. Violations of this principle will result in a grade of zero in the course and additional disciplinary action including expulsion as determined by CURE.

COURSE OBJECTIVES

Upon the successful completion of this courses the student should be able to:

- Describe the cardiac anatomy and associated vasculature.
- Discuss the technical aspects of Fetal echocardiography imaging.
- List the three factors that contribute to congenital heart disease.
- Describe why the four-chamber view cannot rule out all forms of congenital heart disease.
- Scan in accordance with ALARA's Policy on Ultrasound equipment use and safety.
- Understand and successfully perform different fetal echocardiography scanning protocols.
- Recognize normal and pathologic Fetal echocardiographic appearances.
- Describe the characteristics of the spectral doppler waveforms obtained during Fetal echocardiography scanning.
- Discuss the differential diagnosis of abnormal Fetal echocardiographic findings.
- Discuss common errors associated with performing and interpreting Fetal echocardiographic imaging.
- Explain how artifacts can be handled to avoid the pitfalls and misdiagnoses that they can cause.
- Record and save video clips and images of the normal or pathologic Fetal echocardiographic appearance pattern.
- Pass the didactic course with a minimum passing grade of 73.
- Pass the scanning laboratory skills related to echocardiographic ultrasound.

Attendance Policy

Routine, regular class attendance is a requirement for this course.
Promptness and dependability are the soft skills that garner special attention

Attendance Rubric:

Up to 25% will be deducted for each hour missed or for any part thereof (5% for each 10 minutes missed). Students are required to meet their professor to discuss matters of tardiness or absence.

Students are required to notify their instructor via E mail or voicemail to communicate an absence or lateness from class

A student absent from all of his or her classes for a period of 14 consecutive calendar days during which classes or examinations are in session will be regarded as having withdrawn from the program and will receive a grade of W for each course in which he or she is enrolled.

HOMEWORK

An essential part of every program of study, homework enables students to master course learning objectives and leads toward the achievement of the overall program objectives. Homework refers to self-directed learning activities such as reading assignments, research activities and projects, quiz/exam preparation, knowledge checks and self-assessments, etc.

Late Homework/Make-Up Policy

Late homework will be accepted under the following guidelines:

1. It must be turned in within two business days (Monday through Thursday) after the due date.
2. However, homework turned in after the due date will be penalized 10% for each day it is late up to two calendar days after the due date.
3. Homework submitted after three business days after the due date will earn a zero.

Exams and quizzes - Make up of a quiz or an exam will be allowed under the following extenuating circumstances only:

- Documented illness
- Military duty
- Death of a family member
- Court appearance or jury duty

Documentation of the above circumstances must be provided to the Program Director. ***The ability to make-up a quiz or exam must be approved by the Program Director.***

All assignments should be submitted directly to the instructor.

Classroom Policy

Our goal is to build a respectful learning and work environment that allows for positive communication and teamwork. To support your development, all students must abide by academic policies related to attendance, behavior and professional conduct as published in the current institutional catalog. Special attention is granted to the following items:

- There is to be no eating or drinking in any lab/computer class.
- There is to be no eating in ANY class.
- Cellular phones may only be used during class if/when advised by the instructor and required for a learning activity.
- All students must be in full uniform which includes white or black shoes at **ALL TIMES.**
- **There are no rescheduled exams *unless specified by the professor* with advance notification.**
- The determination to accept late homework assignments is made only by the instructor and occurs only in the event of extraordinary circumstances. If the professor does allow for an extension on assignments, there will be an automatic daily reduction by 10 points, each day beyond the original due date.
- All classes are expected to start and end at a predetermined start and end time.

Accommodations for Students with Disabilities

If you have a special learning need, please contact the Program Director. The Program Director and the Campus President will work with the student to identify reasonable accommodations or adjustments necessary to enable him or her to fully participate in the educational processes.

Professional Responsibility

Students are expected to aspire to the same high level of ethics and professionalism that is demanded of registered Ultrasound students. This requires students to conduct themselves as professionals and to fully engage in the learning process by preparing for and participating in all classroom, laboratory and clinical activities.
Attendance is taken each hour of class by the instructor and turned over to the Registrar before the end of the class day. Early departures, tardiest, and class cuts will be recorded in quarter-hour increments and a proportional grade reduction will also be applied. A period of less than 15 minutes will be counted as a quarter-hour of absence. Attendance records are maintained by the Registrar as part of the student's permanent academic record.

INTRODUCTION TO FETAL ECHOCARDIOGRAPHY SYLLABUS

COURSE OUTLINE
Below is the course schedule which identifies lecture topics, assignments and other course activities and the homework required for completion each week.

The course syllabus is a general plan for the course; deviations may be necessary and will be announced.

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week One Objectives: • Introduction to course, class policies, and Syllabus • Describe the landmarks of the thoracic cavity. • Define the relational landmarks of the heart. • Describe the anatomic landmarks of the cardiac chambers, valves and interventricular septum. • How to scan a pediatric patient.		
DAY 1. **Fetal echocardiography: Beyond the four chambers** Risk Factors Indicating Fetal Echocardiography Fetal Risk Factors Maternal Risk Factors Familial Risk Factors Beyond the Four-Chamber View Transducer Requirements Instrumentation Motion Mode Imaging	Power Point Presentation/ Lecture	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 35: Fetal Echocardiography: Beyond the Four Chambers pages 910-926 Chapter 35: Workbook: pages 275-282
DAY 2. Fetal echocardiography: Beyond the four chambers Pulsed Doppler Imaging Color Flow Doppler Imaging Three-Dimensional Imaging Fetal Ultrasound Landmarks Echocardiographic Evaluation of the Fetus Four-Chamber View Left and Right Ventricular Outflow Tracts Ductal and Aortic Arch Views: Oblique Long Axis	Power Point Presentation/ Lecture REVIEW/ FINAL EXAM	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 35: Fetal Echocardiography: Beyond the Four Chambers pages 910-926